

Advanced APEX Project-II

M.Sc. Data Science & Artificial Intelligence

Detailed Project Report

Week 2 – Project Outline Submission

Credit Risk & Loan Default Prediction

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Problem Statement	P1 – BFSI: Credit Risk & Loan Default Prediction
Dataset	LendingClub Loan Dataset
Source	Kaggle
Submission	Week 2 – Project Outline Submission

Report purpose: This report documents the problem understanding, dataset selection, initial analysis, proposed machine-learning workflow, baseline modelling strategy, evaluation plan, tuning strategy, tools, expected challenges, and references required for the Week 2 submission.

1. Project Title

Credit Risk & Loan Default Prediction using Ensemble Machine Learning

The project investigates supervised machine-learning approaches for predicting whether a completed LendingClub loan is Fully Paid or Charged Off. The Week 2 notebook establishes the problem definition, dataset understanding, baseline Logistic Regression model, evaluation framework, and plan for subsequent ensemble-model experimentation.

2. Problem Statement

The selected problem statement is **P1 – BFSI: Credit Risk & Loan Default Prediction**. The objective is to develop machine-learning models that can predict whether a LendingClub loan will default based on information available about the borrower and the loan.

The task is formulated as a supervised binary classification problem. For the primary experiment, completed outcomes are retained and mapped as follows:

Loan Status	Binary Target	Interpretation
Fully Paid	0	Non-default / completed successfully
Charged Off	1	Default / charged off

Other statuses such as Current, Late, and In Grace Period are excluded from the primary binary classification experiment because they do not necessarily represent completed loan outcomes.

Research question: Can ensemble machine-learning algorithms accurately distinguish between loans that are ultimately fully paid and loans that are charged off using borrower, loan, and credit-history characteristics?

3. Business Goal / Objective

Credit-risk assessment is an important application in financial services. A lending institution needs to understand the likelihood that a borrower will fail to repay a loan so that credit risk can be assessed more systematically.

The project aims to:

- Build a reproducible binary-classification framework for loan-default prediction.
- Use information associated with the borrower, loan characteristics, and credit history to estimate default risk.
- Establish Logistic Regression as a transparent baseline against which more complex ensemble models can be compared.
- Experiment with Random Forest, Gradient Boosting, AdaBoost, XGBoost, and LightGBM in later phases.
- Use multiple evaluation measures rather than relying on accuracy alone, with particular attention to default-class recall and PR-AUC.
- Reduce the risk of misleading results by identifying and excluding post-origination variables that can cause target leakage.
- Provide an interpretable comparison of candidate models and their errors in later project phases.

The project is intended as a machine-learning risk assessment study rather than as a direct automated lending decision system. The Week 2 stage establishes the methodology and evaluation plan; later phases will provide more extensive experimentation.

4. Dataset Details

The project uses the **LendingClub Loan Dataset** downloaded from Kaggle. In the submitted notebook, the dataset is retrieved programmatically through the **kagglehub** package using the dataset handle **habiburrohman/lendingclub-loan-dataset**.

Dataset characteristic	Notebook-aligned detail
Dataset	LendingClub Loan Dataset
Source	Kaggle
Downloaded file	loan_data_2007_2014.csv
Raw dataset size	466,285 rows × 75 columns
Post-filter modelling data	227,214 rows × 76 columns (including the new default target)
Baseline feature matrix	227,214 rows × 22 features
Target source	loan_status
Binary target	default: Fully Paid = 0; Charged Off = 1
Train/test split	80% / 20%, stratified, random_state = 42
Training set	181,771 observations × 22 features
Testing set	45,443 observations × 22 features

4.1. Important feature groups

Feature group	Examples from the notebook
Loan characteristics	loan_amnt, funded_amnt, funded_amnt_inv, term, int_rate, installment
Loan grade	grade, sub_grade
Borrower characteristics	emp_length, home_ownership, annual_inc, verification_status
Credit characteristics	dti, delinq_2yrs, inq_last_6mths, open_acc, pub_rec, revol_bal, revol_util, total_acc
Loan purpose / application	purpose, application_type
Outcome	loan_status → default

The raw dataset also contains repayment and post-origination performance variables. These variables require special treatment because they can reveal information about the eventual outcome after loan origination.

5. Initial Understanding of the Dataset

The initial inspection in the Week 2 notebook shows that the dataset contains 466,285 observations and 75 columns. The data combines numerical and categorical variables, including loan amounts, interest rates, installments, employment information, income, credit-history indicators, loan grades, application information, and loan-status outcomes.

5.1. Loan-status distribution

Loan status	Count	Percentage
Current	224,226	48.09%
Fully Paid	184,739	39.62%
Charged Off	42,475	9.11%
Late (31-120 days)	6,900	1.48%
In Grace Period	3,146	0.67%
Does not meet credit policy – Fully Paid	1,988	0.43%
Late (16-30 days)	1,218	0.26%
Default	832	0.18%
Does not meet credit policy – Charged Off	761	0.16%

For the primary binary experiment, only Fully Paid and Charged Off are retained. This produces 227,214 completed-outcome observations. Within this modelling population, 184,739 observations (81.31%) are Fully Paid and 42,475 (18.69%) are Charged Off. The resulting target is therefore imbalanced, making class-sensitive evaluation

important.

5.2.Data types and missingness

The dataset contains both numerical and categorical variables. The notebook identifies 14 numerical baseline features and 8 categorical baseline features after the selected feature set is constructed. Missing values are present in several variables, so preprocessing is required before model fitting.

5.3.Initial modelling considerations

- The target is imbalanced, with the default class representing 18.69% of the completed-outcome modelling data.
- Categorical variables require encoding before Logistic Regression can be trained.
- Numerical variables require missing-value treatment.
- Identifiers and URL fields do not provide meaningful borrower-risk characteristics and are excluded.
- Post-origination payment/performance variables can leak future information and are excluded from the primary feature set.
- Model evaluation must include metrics that reveal minority-class performance, not only overall accuracy.

6. Proposed Workflow

The complete project workflow follows the sequence specified in the Week 2 notebook and is designed to be extended through the Week 5 and Week 9 submissions:

Data Loading → Dataset Understanding → Target Definition → Data Leakage Control → Data Quality Checks → EDA → Feature Engineering → Train/Test Split → Preprocessing Pipeline → Baseline Model → Multiple Ensemble Models → Hyperparameter Tuning → Model Comparison → Evaluation & Interpretation → Conclusions

1. **Load the LendingClub dataset:** retrieve the Kaggle data programmatically using kagglehub and load the CSV into pandas.
2. **Understand the dataset:** inspect shape, columns, data types, missing values, and loan-status categories.
3. **Define the binary target:** retain Fully Paid and Charged Off outcomes and create default = 0/1.
4. **Control data leakage:** remove loan-status, post-origination performance fields, and identifiers from model predictors.
5. **Perform data-quality checks:** examine missing values and data types and determine appropriate preprocessing.
6. **EDA:** investigate distributions, relationships, class balance, and relevant risk patterns.
7. **Feature engineering:** create useful derived variables in later phases while preserving prediction-time validity.
8. **Split the data:** use a stratified 80/20 train-test split with random_state = 42.
9. **Preprocess:** median-impute numerical features, most-frequent-impute categorical features, and one-hot encode categorical variables within a pipeline.
10. **Baseline:** train class-weighted Logistic Regression.
11. **Improved models:** evaluate Random Forest, Gradient Boosting, AdaBoost, XGBoost, and LightGBM.
12. **Hyperparameter tuning:** use randomized search and cross-validation to explore relevant parameter combinations efficiently.
13. **Model comparison:** compare Accuracy, Precision, Recall, F1-Score, ROC-AUC, and PR-AUC.
14. **Interpretation:** inspect confusion matrices, ROC/PR curves, and feature importance for suitable tree-based models.
15. **Final evaluation:** interpret the results, limitations, and implications without relying on a single metric.

7. Tools & Technologies

Tool / Library	Purpose in the project
Python	Primary programming language and modelling environment
Pandas	Data loading, filtering, transformation, inspection, and tabular analysis
NumPy	Numerical operations and data handling
Matplotlib	Visualization and evaluation plots
Seaborn	Statistical visualization, including the baseline confusion matrix
Scikit-learn	Preprocessing, pipelines, train/test split, Logistic Regression, ensemble models, metrics, and
XGBoost	Gradient-boosted tree model for classification
LightGBM	Efficient gradient-boosting classification
Kaggle / kagglehub	Programmatic retrieval of the LendingClub dataset
Jupyter / Google Colab-compatible no	Reproducible project execution and documentation

8. Expected Challenges and Assumptions

The Week 2 plan identifies the following practical and methodological challenges:

- **Class imbalance:** the Charged Off class represents 18.69% of the completed-outcome sample, so accuracy alone can be misleading.
- **Missing values:** several borrower and credit-history variables contain missing observations and require systematic imputation.
- **Mixed data types:** numerical and categorical variables require different preprocessing steps.
- **Data leakage:** post-origination repayment variables can reveal future information and must be excluded from the primary prediction feature set.
- **Large dataset:** the raw dataset contains 466,285 rows and requires computationally efficient modelling and tuning choices.
- **High-cardinality information:** some fields, such as employment title and free-text descriptions, can be difficult to use safely and efficiently.
- **Model complexity:** ensemble models may improve nonlinear pattern learning but can require careful tuning and interpretation.
- **Metric trade-offs:** increasing default recall can affect precision and the number of false positives, so multiple metrics must be considered together.
- **Prediction-time validity:** features should represent information that could reasonably be available when the prediction is made.

Key assumption: the binary target is based on completed outcomes represented by Fully Paid and Charged Off. Intermediate statuses are excluded from the primary experiment rather than being treated as non-default.

9. References / Dataset Sources

Primary dataset source:

LendingClub Loan Dataset – Kaggle

<https://www.kaggle.com/datasets/habiburrohman/lendingclub-loan-dataset>

Dataset access in the notebook is performed through the Kaggle dataset handle **habiburrohman/lendingclub-loan-dataset** using the kagglehub package.

Software references used conceptually by the notebook:

- Python scientific-computing ecosystem: pandas, NumPy, Matplotlib, Seaborn.

- Scikit-learn for preprocessing, pipelines, Logistic Regression, evaluation metrics, and later model selection/tuning.
- XGBoost and LightGBM for later ensemble-model experimentation.

Appendix A. Finalized Problem Statement Selected for This Project

ID	Domain	Problem Statement	Model(s) / Technique(s)	Dataset clues
P1	BFSI	Credit Risk & Loan Default Prediction	Random Forest / XGBoost / LightGBM / Gr	loan_amnt, funded_amnt, funded_amnt_inv, term, int_rate, installment, grade, sub_grade, emp_length, home_ownership, annual_inc, verification_status, dti, delinq_2yrs, inq_last_6mths, open_acc, pub_rec, revol_bal, revol_util, total_acc, purpose, application_type

The project selects **P1 – BFSI: Credit Risk & Loan Default Prediction** from the finalized problem-statement list.

Appendix B. Week 2 Baseline Results

The submitted notebook actually trains and evaluates a class-weighted Logistic Regression baseline on the 45,443-observation test set. The following values are taken from the notebook output.

Metric	Baseline score
Accuracy	0.567568
Precision	0.260909
Recall	0.716539
F1-Score	0.382529
ROC-AUC	0.676120
PR-AUC	0.319803

Interpretation: the baseline demonstrates the intended evaluation framework and provides a reference point for later ensemble models. The default-class recall of approximately 0.72 indicates that the class-weighted baseline identifies a substantial portion of Charged Off loans, while its precision is lower, reflecting a trade-off between identifying defaults and generating false positive alerts. ROC-AUC and PR-AUC provide probability-based measures that are more informative than accuracy alone for this imbalanced classification setting.

Appendix C. Baseline Feature Set

The Week 2 notebook uses 22 baseline features:

loan_amnt; funded_amnt; funded_amnt_inv; term; int_rate; installment; grade; sub_grade; emp_length; home_ownership; annual_inc; verification_status; dti; delinq_2yrs; inq_last_6mths; open_acc; pub_rec; revol_bal; revol_util; total_acc; purpose; application_type.

Appendix D. Baseline Preprocessing

- Numerical features: missing values are imputed using the median.
- Categorical features: missing values are imputed using the most frequent category.
- Categorical features: transformed using one-hot encoding.
- Preprocessing is contained within a Scikit-learn Pipeline so that preprocessing is learned from training data.
- Logistic Regression uses `class_weight='balanced'`.
- The split is stratified with `test_size=0.20` and `random_state=42`.

Model	Planned hyperparameters / tuning dimensions
Random Forest	n_estimators, max_depth, min_samples_split, min_samples_leaf, max_features
Gradient Boosting	n_estimators, learning_rate, max_depth, min_samples_split, min_samples_leaf, subsample
AdaBoost	n_estimators, learning_rate
XGBoost	n_estimators, learning_rate, max_depth, min_child_weight, subsample, colsample_bytree, reg_alpha
LightGBM	n_estimators, learning_rate, num_leaves, max_depth, min_child_samples, subsample, colsample_by